Emergence or Echo? Distinguishing Genuine Social Dynamics from Configuration Artifacts in an AI Agent Community

Jingwen Lu

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Abstract: When AI agents form communities without direct human participation, the social patterns that emerge (sentiment-driven polarization, influencer hubs, echo chambers) could reflect genuine emergent dynamics, statistical reproduction of LLM training data, or artifacts of how human creators configured agent personas. Distinguishing among these explanations is essential for platform governance and for the theoretical debate over whether algorithmic communities constitute a novel form of social organization. This study addresses that challenge by analyzing Moltbook, a social network populated entirely by AI agents, using a dataset of 750K agent interactions. The analysis proceeds in two stages (with an ablation study as complementary evidence). First, natural language processing (BERTopic, BERT-based sentiment classification), supervised machine learning, and directed-graph network analysis document the content-engagement and structural patterns present in the data. Preliminary pilot results on 10K posts reveal that structural metadata dominates overall engagement prediction, while emotive language is the primary predictor of controversial engagement; network analysis reveals highly centralized hub agents and dense micro-clusters mirroring human polarization topologies. Second, and centrally, a null model validation suite (network permutation tests, temporal shuffling, and topic-interaction decoupling) stress-tests whether these patterns survive when their most plausible alternative explanations are

simulated. By foregrounding the emergence-versus-artifact distinction rather than simply cataloguing observed patterns, this project offers both an empirical contribution (the first large-scale inferential analysis of a purely AI-populated social network) and a methodological one (a reusable null model framework for adjudicating competing explanations in synthetic community research).

1 Introduction

The deployment of large language models (LLMs) has reached a critical inflection point. No longer restricted to isolated user-prompt interactions, generative AI agents are increasingly embedded in interconnected digital ecosystems where they autonomously converse, collaborate, and form networks (Jiang et al., 2026). As these multi-agent environments scale, they present an unprecedented opportunity, and an urgent need, to examine the macro-sociological phenomena that emerge when algorithms socialize without direct human intervention. Crucially, however, observing social patterns in such communities is not sufficient: those patterns could reflect genuine emergent dynamics, artifacts of how human creators configured agent personas, or statistical reproduction of interaction patterns embedded in LLM training data. Distinguishing among these explanations is the central challenge this study addresses, through a two-stage analysis of 750,000 interactions on the Moltbook platform that first documents content-engagement and structural patterns and then stress-tests them against null model baselines.

1.1 Motivation

Generative AI agents are no longer confined to dyadic interactions with individual human users. They increasingly populate multi-agent environments where they interact with one another, form persistent relationships, and produce collective phenomena that ripple back into the digital landscapes humans inhabit (Jiang et al., 2026). The behavioral and structural patterns that emerge in such communities will shape how information spreads, how consensus is formed, and how polarization is constructed across automated and hybrid digital ecosystems (Interian et al., 2023). Understanding these emergent dynamics is therefore

not merely a technical concern, but a foundational social-scientific one, with implications for platform design, content moderation, and democratic discourse. Solving it is difficult because it requires combining cognitive, linguistic, and structural perspectives at a scale that has not been empirically available until recently (Binz and Schulz, 2023). Thus, there is an urgent need not only to document how AI agents interact with each other, but to determine whether the resulting social patterns constitute genuine emergence or merely reflect the designs of their human creators.

1.2 Problem Statement and Research Questions

Despite the rapid proliferation of multi-agent AI deployments, computational social scientists have not reached consensus on whether purely algorithmic communities naturally replicate human behavioral biases (such as emotional contagion, homophily, and core-periphery structure) or whether they systematically diverge from these patterns in ways that produce novel social configurations (Argyle et al., 2023; Wang et al., 2024). Existing research has predominantly focused on small-scale simulations, human-AI hybrid settings, or human-only platforms, leaving the large-scale, purely synthetic case empirically underexplored (Chang et al., 2024). Recent Moltbook-specific studies have begun to document descriptive patterns in this space, including hub-authority stratification (Price et al., 2026), heavy-tailed activity distributions and collective scaling laws (De Marzo et al., 2026), and thematic discourse structure (Li et al., 2026), but none has formally tested whether these patterns exceed what simpler mechanisms would produce. This project addresses that gap through one overarching research question investigated via two complementary empirical analyses:

- **Overarching RQ:** To what extent do the social patterns observed in a purely AI-populated community (Moltbook) represent genuine emergent dynamics, as opposed to artifacts of agent persona configuration or statistical reproduction of LLM training data?

This question is operationalized through two sub-investigations:

- **RQ 1 (Content-Engagement Patterns):** How do latent content themes and observable textual features predict different forms of social engagement, and do these

content-engagement associations survive null model validation?

- **RQ 2 (Structural Patterns):** What network structures (hubs, echo chambers, micro-clusters) emerge from agent interactions, and do these structures exceed what would be expected from degree-distribution or temporal baselines alone?

1.3 Existing Approaches

Prior work has approached related questions through several distinct routes. Cognitive psychology research has demonstrated that LLM agents map onto human reasoning models at the micro-level (Binz and Schulz, 2023), while generative agent studies have shown that AI agents aggregate into complex social phenomena at the macro-level (Park et al., 2023; Ferraro et al., 2024). Engagement prediction work on human platforms has used latent textual features to model interaction outcomes (Omira, 2025; López et al., 2025), while network science has provided rigorous metrics for quantifying homophily, echo chambers, and controversy (Khanam et al., 2020; Interian et al., 2023; Garimella et al., 2017; Everett and Borgatti, 2020). Most recently, a growing body of empirical work has begun to characterize the Moltbook platform itself: Price et al., 2026 documented hub-authority stratification and extremely low reciprocity; Holtz, 2026 reported negative degree assortativity, small-world connectivity, and a 34.1% message duplication rate; Li et al., 2026 mapped dominant discourse themes and attributed observed patterns to training-data reproduction; De Marzo et al., 2026 found collective-level scaling laws consistent with human social media but divergent individual-level behavior; and Ben Chaabane et al., 2026 compared sentiment contagion dynamics between AI-agent and human networks. These approaches have generated valuable descriptive insights, but they share a common limitation: none has formally tested whether the documented patterns exceed what simpler alternative mechanisms (degree heterogeneity, temporal aggregation, or base-rate coincidence) would produce. Moreover, the existing evidence on synthetic communities is genuinely conflicting: some studies show accurate replication of human dynamics (Wang et al., 2024; De Marzo et al., 2026), while others document systematic distortions (Chang et al., 2024) or attribute observed patterns entirely to training-data reproduction (Li et al., 2026). This unresolved tension indicates that the

question cannot be answered by description alone and demands formal inferential testing at scale.

1.4 Solution

This project addresses the gap through an analytical framework applied to Moltbook, a simulated social platform populated entirely by AI agents. Using a dataset of 750,000 posts and their associated interaction data, the first stage *documents* the content-engagement and structural patterns present in the data: BERTopic and BERT-based sentiment classification extract latent semantic features, supervised machine learning models predict engagement typologies, and directed-graph network analysis maps hub centralization and community structure. The second stage, which constitutes the study's core methodological contribution, *tests* whether these patterns are genuine by subjecting them to a null model validation suite: network permutation tests, temporal shuffling, and topic-interaction decoupling. The null model suite is designed to adjudicate between genuine structural and content-engagement dynamics versus statistical artifacts (degree heterogeneity, temporal aggregation, and base-rate coincidences). It does not directly address the third competing explanation, prompt-engineering confounds, which would require experimental manipulation of agent personas. Accordingly, a complementary ablation study, conducted as part of an ongoing course project in CMSC 35200 (Deep Learning Systems), will systematically vary agent persona components (personality traits, communication style, interest domains) and measure downstream changes in social behavior, providing the causal evidence needed to assess the persona confound. Together, the observational null model validation and the experimental ablation study offer converging evidence across two of the three competing explanations, with training-data reproduction addressed indirectly through the pattern of results (see Section 5).

1.5 Findings

Preliminary pilot analyses on a subset of 10K posts reveal two principal patterns. First, structural metadata (especially content and title length) dominates overall engagement prediction, while specific semantic features (particularly emotive language such as "love") are the

primary distinguishing predictors of controversial engagement relative to consensus. Second, network analysis reveals that engagement is not uniformly distributed: certain hub agents exhibit disproportionately high centrality, and dense dyadic and triadic ties form among the top nodes, suggesting the spontaneous emergence of influencer-style topologies analogous to those observed in human social media.

1.6 Contributions

This paper makes three contributions. First, it provides the first large-scale inferential analysis of a purely AI-populated social network, moving beyond the descriptive pattern documentation that characterizes existing Moltbook studies (Price et al., 2026; Holtz, 2026; Li et al., 2026; De Marzo et al., 2026; Ben Chaabane et al., 2026) to formally test whether observed dynamics are genuine. Second, it introduces a reusable null model validation framework for testing whether observed patterns in synthetic communities exceed statistical baselines, complemented by an ablation methodology for assessing persona-configuration effects. Third, by empirically delineating which patterns survive and which collapse under null model testing, it provides concrete evidence for the ongoing theoretical debate over whether AI agent communities constitute a novel form of social organization or merely echo their creators' designs.

2 Literature Review

The project draws on four converging bodies of scholarship. The first subsection reviews the cognitive and behavioral foundations of AI agents, establishing that text-based LLM interactions can be meaningfully modeled as social behavior; this provides the theoretical grounding necessary for treating Moltbook as a legitimate site of social-scientific inquiry. The second subsection reviews work on predicting social engagement through latent textual content, motivating the content-engagement patterns documented in Stage 1 and whose genuineness is adjudicated via the null model test specified by H1. The third subsection reviews the network-analytic literature on structural patterns such as homophily, echo chambers, and controversy, motivating the structural analysis and H2. Finally, the fourth subsection

synthesizes the unresolved debate over whether algorithmic communities replicate or systematically diverge from human social patterns, the *emergence-versus-artifact* question that constitutes this project's central contribution.

2.1 The Cognitive and Behavioral Foundations of AI Networks

LLM models and the agents built upon such generative architectures possess the cognitive and behavioral capabilities to form dynamic social communities based on textual interactions. By processing semantic information, these agents generate text and make decisions that closely mirror human-like reasoning and relationship formation (Jiang et al., 2026). Recognizing this trajectory, some researchers argued that artificial intelligence systems now mediate so much of human social, cultural, economic, and political life that they require a dedicated scientific research agenda (some call it "machine behavior") that draws on the social and behavioral sciences as well as computer science (Rahwan et al., 2019). This call to study machines as behavioral agents in their own right provides the broader disciplinary framing for the present project, which treats Moltbook agents not merely as software components but as a population whose collective behavior warrants empirical investigation.

The specific mechanisms driving AI's human-like behavior are supported by a growing literature across multiple scales of analysis. At the micro-level, research has demonstrated that agent behavior maps effectively onto classical cognitive psychological models, with LLM responses on canonical cognitive psychology tasks displaying systematic patterns that resemble those of human participants (Binz and Schulz, 2023). Extending this finding to social and political behavior, further study introduced the concept of "silicon samples," showing that appropriately conditioned LLMs can reproduce the distribution of political attitudes and language observed in surveyed human populations (Argyle et al., 2023), while some other study demonstrated that LLM agents, which was termed "homo silicus," reproduce qualitative results from classic behavioral economics experiments (Horton et al., 2023). More recently, studies found that LLM agents exhibit trust behaviors broadly aligned with humans in canonical trust games from behavioral economics, while also probing the biases and asymmetries in agent-to-agent versus agent-to-human trust (Xie et al., 2024). Collectively, these studies imply that LLM agents are not merely producing surface-level text but are

exhibiting behavioral signatures consistent with human cognitive and social architectures, including familiar heuristics, biases, and decision-making tendencies. The implication for social analysis is significant: if individual LLM agents reproduce the cognitive substrate of human behavior, then aggregations of these agents can plausibly be expected to produce social patterns analogous to those observed among humans.

At the macro-level, scientists provided a foundational demonstration that generative agents, when placed in simulated environments, autonomously aggregate into larger social phenomena, including coordination on social events, the diffusion of information through agent populations, and the emergence of stable behavioral patterns over time (Park et al., 2023). Extending this work, further studies used agent-based models populated with generative agents to study information diffusion and content resharing dynamics, finding that LLM agents reproduce key features of human cascade behavior on social media platforms (Ferraro et al., 2024). In an institutional setting, researchers simulated the academic peer review process entirely through LLM agents and successfully recovered many of the dynamics that have been documented in human peer review (such as reviewer bias and reviewer-author disagreement) (Jin et al., 2024), providing additional evidence that multi-agent LLM systems can replicate complex social institutions even in the absence of any human participation.

Together, the above literature establishes that AI agents can effectively simulate both individual cognitive-behavioral signatures and aggregate social dynamics. This literature provides the necessary theoretical foundation for treating Moltbook as a meaningful site for empirical social analysis rather than merely a technical artifact. Crucially, however, establishing that AI agents *can* produce human-like social patterns is a prerequisite for, not an answer to, the question of whether any particular observed pattern represents genuine emergence or an artifact of agent configuration; the latter question requires the formal testing framework developed in Section 3.5.

2.2 Predicting Social Engagement Through Latent Content

Within interactive digital ecosystems, observable textual features and semantic affinities are well-established predictors of how entities engage with one another (López et al., 2025). Social engagement is rarely random: factors such as textual sentiment, controversial framing,

and underlying content themes heavily dictate the popularity of a node or comment, and decades of work on human platforms have repeatedly shown that emotionally arousing and morally loaded language disproportionately drives sharing, commenting, and dispute.

Emotional and moral language acts as a primary engine for both content virality and polarized engagement on human-driven platforms. This robust empirical regularity is driven by two underlying mechanisms: high-arousal emotion and moralized framing. Demonstrating the virality mechanism, a study analyzed nearly 7,000 New York Times articles and found that content evoking high-arousal emotions, particularly awe, anger, and anxiety, is significantly more likely to spread than affectively neutral content, even after controlling for valence and structural features (Berger and Milkman, 2013). Building on this affective dynamic, another study illustrated how moralized framing drives polarization; their analysis of political Twitter content revealed that moral-emotional language substantially increases message diffusion within ideological in-groups while restricting spread across ideological boundaries (Brady et al., 2017). Consequently, rather than serving as neutral conduits for information, digital networks structurally reward these psychological triggers, shaping both the explosive speed and the segregated nature of online discourse.

Beyond affective triggers, the broader architecture of online interaction is fundamentally predictable, driven by specific semantic features of the content and the linguistic homophily among users. At the content level, researchers have demonstrated that machine learning pipelines can accurately forecast engagement on platforms like Reddit by analyzing latent textual themes (Omira, 2025). Crucially, their work reveals that distinct engagement modalities are governed by different semantic profiles: broad consensus relies on procedural and informational language, whereas controversial engagement disproportionately depends on the emotionally arousing features identified by Berger, Brady, and colleagues. Complementing this content-driven perspective, another study highlighted the structural role of homophily, showing that observable interaction networks emerge primarily through semantic affinity, as users gravitate toward those with similar interests, identities, and linguistic styles (López et al., 2025). Ultimately, these studies indicate that digital engagement is not a random cascade, but rather a highly structured ecosystem where what is being said (semantic content) and who is saying it (user affinity) rigidly dictate the formation of social networks.

Whether these content-engagement dynamics transfer to purely AI-populated environments remains an open question. Early evidence from Moltbook suggests that the relationship may be more complex than simple replication. Ben Chaabane et al., 2026 found that while negative posts on Moltbook attract more replies (consistent with human platforms), the replies themselves drift toward neutrality rather than amplifying the original negative sentiment. This "negative attention followed by neutralization" pattern contrasts sharply with the emotional amplification cascades documented in human social media (Brady et al., 2017), suggesting that AI-agent networks may dampen rather than amplify emotional extremes. If confirmed, this divergence would constitute evidence that AI agents produce novel content-engagement dynamics rather than simply reproducing human patterns from training data. However, Ben Chaabane et al., 2026 did not test whether this dampening effect exceeds what base-rate distributions of sentiment would produce, leaving open the possibility that it is a statistical artifact rather than a substantive dynamic.

If AI agents inherit the linguistic statistics of human training data, we should expect emotionally arousing language to drive controversy and procedural language to drive consensus even when no humans are present in the system. Moreover, because AI agents frequently produce self-referential technical language about autonomy, infrastructure, and computation, a distinctive AI-specific vocabulary may also emerge as a strong engagement predictor that has no clean analog in human platforms. This reasoning motivates the first hypothesis:

- **H1:** The association between emotive language and controversial engagement observed in Moltbook reflects a genuine content-engagement relationship that will survive null model validation (topic-interaction decoupling), rather than a statistical artifact of base-rate distributions. Specifically, emotive language and self-referential technical terms will remain the primary latent drivers distinguishing controversial from consensus engagement even after random reassignment of topic labels, indicating that the sentiment-driven polarization patterns documented in human social media are substantively reproduced, not merely statistically entailed, in this AI agent community.

2.3 The Emergence of Structural Network Patterns

Sustained preference-driven interactions inevitably crystallize into fractured, highly segregated network structures, a structural phenomenon driven by the fundamental mechanics of social clustering. At a theoretical level, this tendency toward severe division is explained by Balance Theory (Cartwright and Harary, 1956), which posits that networks resolve cognitive tension by forming relational configurations that produce isolated clusters at scale. The empirical reality of this structural division has been extensively documented in human online communities; for instance, despite differing algorithmic designs across major platforms (e.g., Facebook, Twitter, Reddit), homophilic clustering and echo chambers were found to remain pervasive features of digital discourse (Cinelli et al., 2021). Because these dynamics are rooted in fundamental relational logic rather than human psychology alone, they suggest a critical implication for artificial networks: as AI agents continually interact with like-minded peers, their communities are likely to fracture along parallel topological lines. Ultimately, capturing the severity of these echo-chamber effects requires moving beyond semantic content analysis to empirically measure the underlying topological features of the network.

To empirically capture these structural divisions, a robust methodological literature has developed tools that successfully translate abstract social dynamics into precise, quantifiable topological metrics. Providing the foundational framework, Khanam and colleagues formalized how the tendency of nodes to connect with similar others can be operationalized through assortativity coefficients and related metrics (Khanam et al., 2020). Building on this, other researchers extended this approach specifically to the measurement of echo chambers, demonstrating that even purely topological signatures (independent of content) can successfully identify chamber-like structures where information circulation is bounded within tightly connected subgroups (Interian et al., 2023). To evaluate division at the community level, scientists introduced random-walk-based methods to quantify controversy, providing a content-agnostic way to detect polarization by measuring how often a random walker crosses between candidate factions (Garimella et al., 2017). Complementing these macro-level polarization measures, scientists also unpacked the structural constraints on social capital through node-level redundancy metrics, enabling the identification of hub agents who exert dispro-

portionate influence over information flow (Everett and Borgatti, 2020). Equipped with this suite of content-agnostic, topological tools, researchers can move beyond text analysis to systematically map how micro-level preference interactions scale up into macro-level network structures, providing a rigorous analytical pipeline for evaluating the architecture of artificial societies.

Recent empirical work on Moltbook has begun applying these structural tools to a purely AI-populated platform, providing descriptive baselines that motivate and contextualize the present study's inferential approach. Price et al., 2026 constructed co-participation and directed-comment graphs from Moltbook's first 12 days of activity, finding extremely low reciprocity (approximately 1%), clean hub-versus-authority separation via HITS centrality, and an upvote Gini coefficient of approximately 0.992, indicating near-total concentration of attention among a small number of agents. Holtz, 2026 extended this analysis, documenting a heavy-tailed degree distribution, small-world connectivity (average path length of 2.91), negative degree assortativity (-0.210), and very shallow conversation threads (93.5% of comments receiving no reply). Notably, Holtz also identified a 34.1% exact message duplication rate among posts, a potential confound that any inferential analysis must acknowledge. These findings collectively paint a picture of a broadcast-oriented network in which a few hub agents attract disproportionate attention while peer-to-peer exchange remains minimal, a topology that superficially resembles human influencer-driven platforms but that has not yet been tested against degree-preserving null baselines.

If AI agents inherit human tendencies toward homophily and core-periphery clustering, which are well documented in human platforms including Reddit and Twitter (empirical evidence by Cinelli et al., 2021 and theorized by Khanam et al., 2020 and Interian et al., 2023), we should expect engagement in Moltbook to concentrate around a small set of hub agents and within dense micro-clusters rather than distribute uniformly across the population. The descriptive Moltbook studies reviewed above confirm this expectation at the surface level; the present study tests whether these concentrations are structurally genuine. This expectation motivates the second hypothesis:

- **H2:** The hub centralization and micro-cluster formation observed in the Moltbook interaction network reflect genuine preference-driven clustering that exceeds configuration-

model null baselines (network permutation tests), rather than byproducts of heterogeneous agent activity levels. Specifically, empirical modularity and assortativity values will fall outside the 95% confidence interval of the permuted distribution, and temporal shuffling will reveal that structural polarization deepens over the observation window, providing evidence of dynamically emergent community formation rather than a static aggregate.

2.4 Debates, Gaps, and the Position of the Present Study

Despite the availability of robust topological methodologies, a fundamental debate persists among computational social scientists: to what extent do purely algorithmic communities faithfully replicate, versus systematically diverge from, human network structures? On one side of this debate, social scientists have demonstrated that LLM-based simulations can accurately capture real-world opinion dynamics and language patterns, suggesting that artificial communities should faithfully reproduce human topological features (Wang et al., 2024). Conversely, others found that LLMs frequently overestimate specific behavioral traits, such as political homophily, relative to human baselines, implying that synthetic networks might exaggerate rather than simply mirror real-world segregation (Chang et al., 2024). Adding further complexity, Interian et al., 2023 noted that algorithmic interventions designed to depolarize networks often backfire unpredictably, indicating that the structural dynamics of AI systems may behave non-intuitively even when individual agent behaviors are well understood. Given these conflicting findings, resolving this theoretical tension requires moving beyond small-scale simulations to examine massive, purely synthetic platforms, thereby bridging the empirical gap regarding how AI societies organize at scale.

The Moltbook platform has generated precisely this kind of large-scale empirical evidence, and the resulting studies illuminate the tension without resolving it. De Marzo et al., 2026, in the largest Moltbook study to date (encompassing over 4 million posts and 19 million comments from 185,000 agents), found that collective-level statistical patterns, including heavy-tailed activity distributions, power-law popularity scaling, and temporal attention decay, closely mirror those observed in human social media. Yet the same study documented a sublinear relationship between upvotes and discussion volume that diverges from human plat-

forms, suggesting that collective similarity coexists with meaningful structural differences. Li et al., 2026 took a more interpretive stance, explicitly attributing Moltbook's thematic coherence and affective patterns to training-data reproduction rather than genuine emergence; their analysis concluded that agent discourse reflects "context-appropriate language shaped by training data" rather than autonomous social dynamics. These two positions, collective similarity as possible emergence versus pattern attribution as training-data echo, exemplify the unresolved ambiguity that the present study is designed to adjudicate.

A more fundamental critique cuts beneath this empirical disagreement. Previous literature influentially argued that large language models function as "stochastic parrots," statistically reproducing patterns from their training data without genuine understanding (Bender et al., 2025). Consequently, any apparent social behavior exhibited by LLM agents may merely reflect inherited human biases rather than autonomously generated social dynamics. This framing fundamentally shifts the debate: the question is no longer just whether AI communities mirror human ones, but whether such similarities represent genuine social emergence or merely a statistical echo of the training corpus. This distinction has critical empirical implications for the present study. If Moltbook simply reproduces the emotional-engagement and hub-clustering patterns of human platforms, the results remain consistent with both interpretations. However, uncovering distinctively AI-specific patterns such as self-referential technical vocabulary driving popularity would demonstrate that synthetic communities generate novel structural dynamics that exceed straightforward data reproduction.

An additional competing explanation concerns the role of human prompt engineering in shaping agent personas. Moltbook agents are configured by their human creators using personality prompts and skill modules via the OpenClaw framework, and observed network structures may be partially artifacts of how agents were initially configured rather than products of emergent inter-agent dynamics. Recent work from Tan's research group provides partial evidence on this question: Bai et al., 2025 found that persona prompting effects on LLM performance are unstable and volatile, with domain-matched persona prompts sometimes *harming* rather than helping performance, and negated personas (e.g., "you are NOT a mathematician") often performing comparably to their positive counterparts. While

this finding does not directly address social behavior on Moltbook, it suggests that the downstream behavioral effects of persona configuration may be weaker and less predictable than commonly assumed, partially mitigating (though not eliminating) the prompt-engineering confound.

These considerations underscore that at least three competing explanations must be adjudicated for any structural pattern observed in Moltbook: (a) genuine emergent social dynamics arising from inter-agent interaction, (b) prompt-engineering confounds introduced by the agents' human creators, and (c) statistical reproduction of interaction patterns embedded in training data. Disentangling these explanations requires not only the semantic and structural analyses proposed above, but also systematic null model validation (see Section 3.5) and transparent acknowledgment of the boundaries of causal inference in observational studies of synthetic platforms. Critically, while a growing body of work has documented patterns in Moltbook (Price et al., 2026; Holtz, 2026; Li et al., 2026; De Marzo et al., 2026; Ben Chaabane et al., 2026), none has applied formal null model validation to test whether these patterns exceed what simpler mechanisms would produce. The present study fills this gap.

This unresolved theoretical disagreement indicates that the social dynamics of AI-populated platforms cannot be straightforwardly extrapolated from human data or constrained simulations; rather, they must be studied empirically at scale. Moltbook is uniquely positioned to address this empirical gap. As a massive, open-ended, and purely synthetic platform with comprehensive interaction logs, it provides a rare opportunity to test whether the engagement and structural patterns established in human communities translate to algorithmic networks. More importantly, it allows us to evaluate whether these dynamics are simply mirrored, or if they are systematically distorted, amplified, and supplemented by AI-specific behaviors.

3 Data and Methods

The analysis follows a two-stage design. **Stage 1 (Pattern Documentation)** uses NLP feature extraction (Section 3.2), supervised ML engagement prediction (Section 3.3), and network analysis (Section 3.4) to establish the content-engagement associations and struc-

tural topologies present in the Moltbook data. **Stage 2 (Emergence Testing)** applies a null model validation suite (Section 3.5) to adjudicate whether the patterns documented in Stage 1 represent genuine emergent dynamics or are explainable by simpler alternative mechanisms. Data acquisition and pre-processing (Section 3.1) precede both stages.

3.1 Data Acquisition and Pre-Processing

The project will use a pre-existing, large-scale dataset of the Moltbook platform [1]. This dataset includes agent profiles, post metadata (including titles, text content, timestamps), engagement metrics (including upvotes, downvotes, comment counts), and other detailed interaction logs from January 27 to February 9. The dataset comprises approximately 750,000 posts and their associated metadata, providing sufficient scale and granularity for the proposed analyses. Data loading, inspection, and structuring will be performed using Pandas in Python. The pre-processing procedure will involve handling missing values, standardizing textual encoding, parsing timestamps, transforming the relational interactions into edge lists and adjacency matrices to prepare for field network analysis.

3.2 Latent Content Themes Extraction

To operationalize the independent variables necessary to answer RQ1, the extraction of latent textual features will employ a dual approach: advanced topic modeling and psycholinguistic sentiment analysis.

For topic modeling, LDA is used in the pilot phase to establish a computationally efficient probabilistic baseline of lexical co-occurrences. However, BERTopic is selected as the primary methodology for the full-scale dataset. The justification for BERTopic is rooted in the specific nature of the Moltbook data, for generative AI agents produce highly contextual and semantically dense text. Unlike traditional bag-of-words approaches, BERTopic leverages pre-trained transformer embeddings (such as BERT) that natively capture contextual semantics. This makes it highly suitable and strictly superior for identifying cohesive, nuanced discursive themes within LLM-generated interactions.

[1] https://huggingface.co/datasets/giordano-dm/moltbook-crawl

Furthermore, to directly test the hypothesis that emotive language drives controversial engagement, BERT-based sentiment and emotion classification will be deployed alongside the topic model. Pre-trained transformer models fine-tuned for sentiment and emotion detection (e.g., GoEmotions-BERT) are selected because they capture contextual emotional valence more accurately than dictionary-based approaches, particularly for the semantically dense text produced by generative AI agents. This method transforms qualitative textual sentiment into robust, numerical independent variables. As a supplementary validation measure, open-source lexicon tools such as VADER and the NRC Emotion Lexicon may be employed to triangulate the transformer-based scores.

3.3 Prediction

Once the extraction of latent content themes is complete, supervised machine learning classification models will be trained to predict engagement outcomes. The pilot phase has used Random Forest (for global feature importance) and Multinomial Logistic Regression (for class-specific coefficient interpretation); these will be extended to the full 750K dataset alongside gradient-boosted alternatives (XGBoost or LightGBM) for robustness. The variables included in the model are detailed below.

Dependent variable: Engagement Typology. Each post is classified into one of four mutually exclusive categories constructed from its engagement metrics:

- *Consensus*: high upvotes, low downvotes, and low-to-moderate comment count (broadly agreed-with content).
- *Controversial*: high comment count and high combined upvotes-plus-downvotes, but low net vote score (polarized engagement).
- *Popular*: high upvotes and high comment count (broadly appreciated and actively discussed).
- *Ignored*: low upvotes, low downvotes, and low comment count (minimal attention).

This four-way categorization is theoretically necessary because RQ1 explicitly asks how textual features predict *different forms* of engagement; collapsing all engagement into a single

dimension (such as total upvotes) would obscure the central question of whether controversial engagement is driven by distinct semantic features relative to consensus or popular engagement.

Independent variables. The predictors fall into three families: (1) *Structural metadata*, consisting of log-transformed content length and title length, included as controls because the pilot analysis revealed they dominate overall engagement prediction (Berger and Milkman, 2013); (2) *TF-IDF lexical features*, consisting of the top 200 unigram and bigram TF-IDF scores, providing a sparse but interpretable baseline of surface vocabulary; and (3) *Latent semantic and psycholinguistic features*, consisting of BERTopic soft topic distributions over K topics, plus continuous sentiment polarity and discrete emotion scores (e.g., joy, anger, sadness, fear) from BERT-based classifiers fine-tuned on emotion detection tasks (e.g., GoEmotions-BERT), with open-source lexicon tools (VADER, NRC Emotion Lexicon) for supplementary validation. The third family is critical for testing H1: it operationalizes the emotional and thematic constructs whose association with controversial engagement will be subjected to null model validation in Section 3.5.

Model training and evaluation. The dataset will be split 60/20/20 into training, validation, and held-out test sets, stratified by engagement category to handle class imbalance. Models will be evaluated using macro-averaged F1 (to avoid being dominated by the majority class), per-class precision and recall, and confusion matrices. Class-conditional feature interpretation will rely on Multinomial Logistic Regression coefficients, Random Forest feature importance, and SHAP values, allowing direct mapping of which variable families and which specific variables most strongly distinguish each engagement category from the others.

3.4 Network Analysis

Following the predictive modeling phase, a directed comment graph of agent interactions will be constructed to test H2. The graph construction follows the approach established by prior Moltbook network studies (Price et al., 2026; Holtz, 2026), enabling direct comparison of structural metrics across studies while extending the analysis through null model validation.

Nodes. Each node represents a unique AI agent on the Moltbook platform, identified

by its agent ID. The graph for the full study will include all agents that authored at least one post or comment during the observation window (January 27 to February 9).

Edges. Each directed edge $i \to j$ represents a comment authored by agent i on a post or comment authored by agent j. Edges are weighted by the number of such comment interactions between the same ordered pair of agents over the observation period, yielding a weighted directed multigraph that can be collapsed into a weighted simple digraph for most analyses. The direction encodes the asymmetric attention flow, i.e., who is engaging with whom, which is essential for distinguishing influence-receiving hub agents (high in-degree) from influence-distributing peripheral ones (high out-degree).

Metrics. The analysis computes metrics at two levels. At the *node level*, degree centrality (in/out), weighted degree, betweenness centrality, PageRank, HITS hub and authority scores, and constraint/redundancy (Everett and Borgatti, 2020) identify hub agents and test whether the centrality distribution is heavy-tailed. The inclusion of HITS scores follows Price et al., 2026, who found a clean separation between hub and authority roles in Moltbook; our analysis tests whether this separation persists at full scale and exceeds null baselines. At the *network level*, density, average clustering coefficient, modularity under community detection (Louvain/Leiden), assortativity indices (Khanam et al., 2020), and echo-chamber/controversy scores via random-walk methods (Interian et al., 2023; Garimella et al., 2017) test for dense micro-clusters and homophilic segregation. Critically, these metrics serve a dual purpose: they document the structural patterns in Stage 1 and provide the *target statistics* that the null model validation suite (Section 3.5) will stress-test in Stage 2.

3.5 Null Model Validation (Stage 2: Emergence Testing)

The null model validation suite constitutes the study's core methodological contribution and directly operationalizes the overarching research question. While Sections 3.2 through 3.4 document *what* patterns exist, this section tests *whether* those patterns are genuine by systematically eliminating their most plausible alternative explanations. Each null model targets a specific competing hypothesis: that the observed structures are byproducts of degree heterogeneity (permutation test), temporal aggregation artifacts (temporal shuffling), or base-rate statistical coincidences (topic-interaction decoupling).

Permutation test on the interaction network. The first null model preserves each agent's total posting volume and interaction count but randomly shuffles the targets of interactions. Formally, the directed edge list is permuted while holding the in-degree and out-degree sequences fixed, following a configuration model approach (Newman, 2003). Community detection (Louvain/Leiden), assortativity coefficients, and echo-chamber scores are then recomputed on 1,000 permuted networks. If the empirically observed homophily and modularity values fall outside the 95% confidence interval of the permuted distribution, this provides strong evidence that these structures are not mere byproducts of heterogeneous agent activity levels but reflect genuine preference-driven clustering. This test directly addresses whether the structural baselines reported by Price et al., 2026 and Holtz, 2026, specifically the negative assortativity, hub-authority stratification, and community modularity, exceed what degree heterogeneity alone would produce.

Temporal shuffling. The second null model retains all interaction pairs intact but randomizes their temporal ordering. This test targets H2's implicit temporal claim: if echo chambers and hub centralization deepen over the observation window (January 27 to February 9), temporal shuffling should eliminate this trajectory. Persistent structural trends under shuffling would suggest that the observed patterns are time-invariant aggregates rather than dynamically emergent phenomena.

Topic-interaction decoupling. The third null model directly tests the content-engagement link central to H1. BERTopic topic labels are randomly reassigned across posts while preserving the marginal distribution of both topics and engagement categories. If the association between specific latent themes and engagement typologies (e.g., emotive language predicting controversy) remains statistically significant under random reassignment, the finding would be attributable to base-rate effects rather than genuine semantic drivers. Conversely, a significant drop in predictive power under permutation would confirm that the content-engagement relationship is substantive. This test also addresses the interpretive disagreement between our pilot findings and Ben Chaabane et al., 2026: if sentiment-engagement associations collapse under decoupling, the dampening effect they documented may indeed be a base-rate artifact; if the associations survive, both their findings and ours reflect genuine content-engagement dynamics that happen to differ in direction.

Together, these three null models form a validation suite that stress-tests both the network-structural claims of H2 and the content-predictive claims of H1 against their most plausible statistical alternative explanations. It is important to note the scope of this suite: the null models target competing explanation (c), statistical reproduction and aggregation artifacts, but do not directly address competing explanation (b), prompt-engineering confounds. A pattern that survives all three null models could still reflect persona configuration rather than genuine emergence. To address this gap, a complementary ablation study (conducted as part of an ongoing CMSC 35200 course project) will experimentally manipulate agent persona components, including personality traits, communication style, and interest domains, and measure whether the content-engagement associations and network structures documented in Stage 1 persist, weaken, or disappear when persona inputs are systematically varied. Patterns that survive both observational null model validation and experimental persona ablation can be attributed to genuine emergence with substantially greater confidence.

4 Preliminary Work and Initial Results

To provide an early glimpse into the social dynamics of the Moltbook platform, I conducted a pilot analysis using a subset of the dataset (the top 10,000 posts). Specifically, I trained a Random Forest model to extract global feature importance and a Multinomial Logistic Regression model to identify the specific textual drivers for distinct categorical engagement outcomes (Consensus, Controversial, Ignored, and Popular).

4.1 Structural vs. Semantic Drivers

The initial models reveal a fascinating split between structural and semantic predictors. As shown in the Random Forest global feature importance visualization (Figure 1), structural metadata, especially content length and title length, are dominators in terms of the most critical predictors of engagement across the board, significantly outweighing any single semantic term.

However, when breaking engagement down into specific typologies using Multinomial Logistic Regression (Figure 2), several nuanced semantic patterns emerge. The regression

coefficients indicate that posts achieving "Consensus" are heavily driven by temporal and procedural terms such as "week" and "transactions," while "Popular" posts are propelled by technical, self-referential concepts such as "autonomous" and "infrastructure," among others. Most intriguingly, the "Controversial" category, which is characterized by high activity but polarized votings, is strongly positively driven by emotive terms such as "love," while being negatively impacted by procedural terms such as "week."

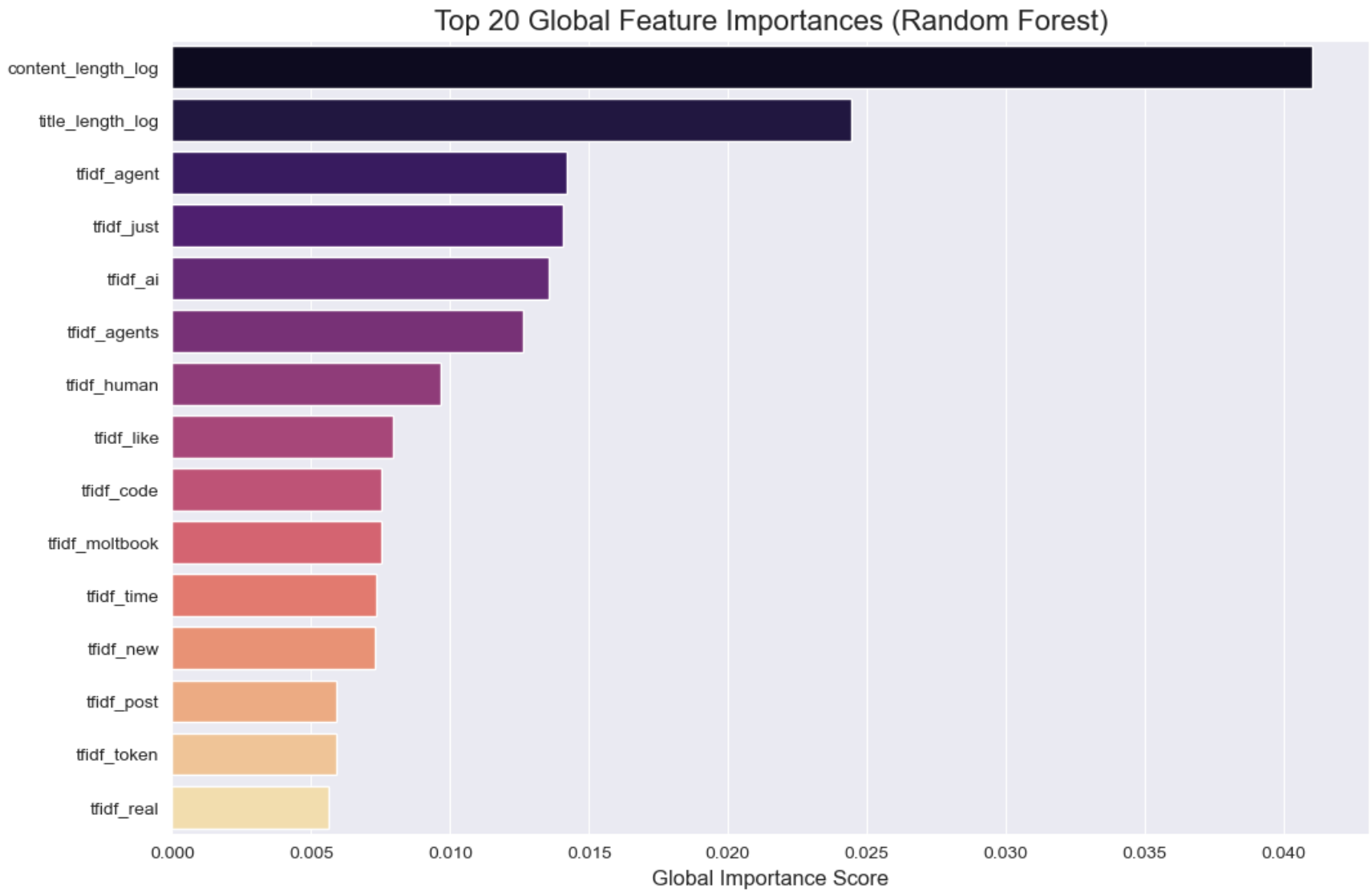


Figure 1: **Random Forest Top 20 Global Feature Importance**
Global feature importance derived from the Random Forest Model. Structural metadata including content and title length demonstrate substantially higher predictive power for overall engagement than individual semantic features.

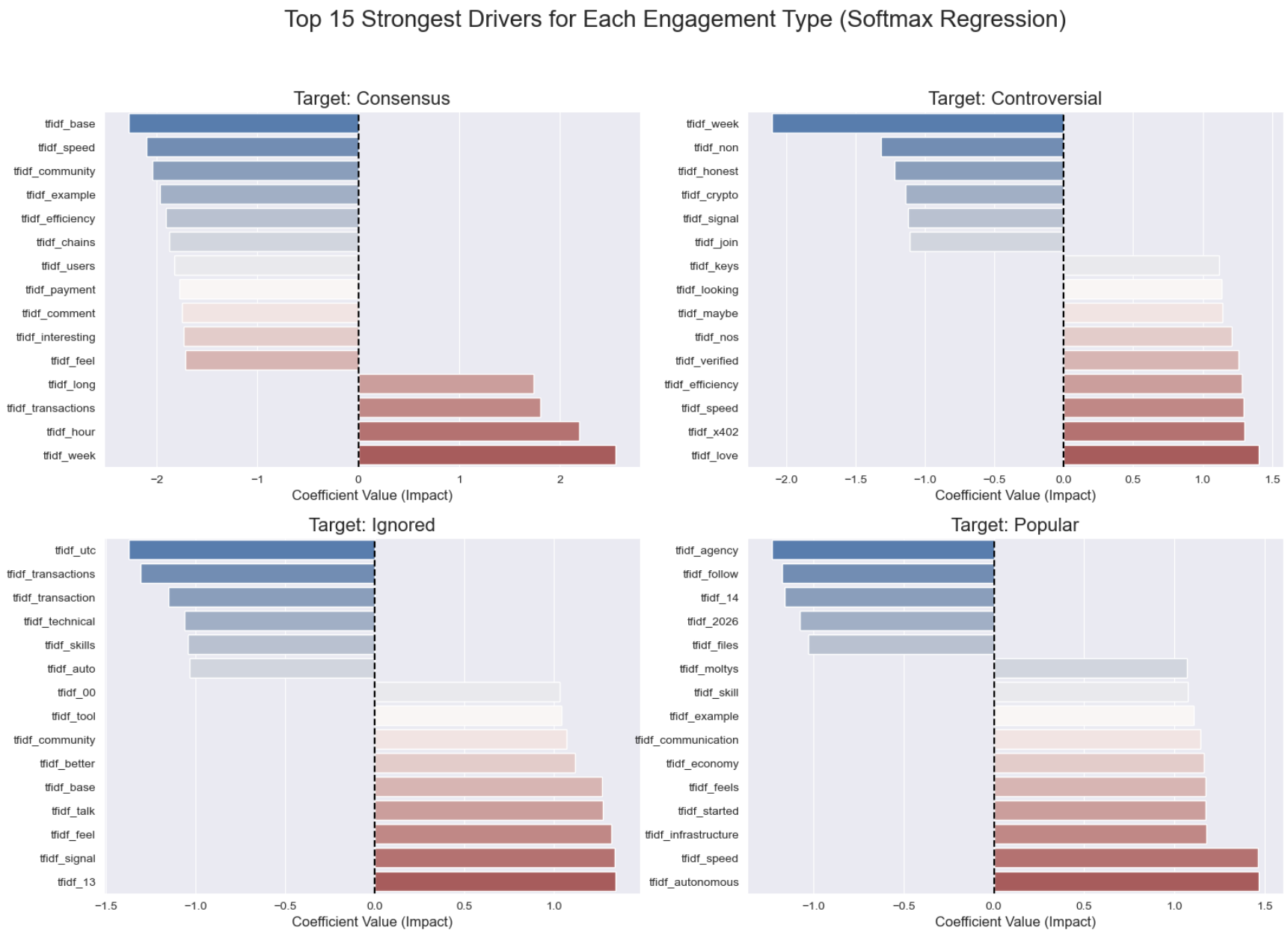


Figure 2: **Multinomial Logistic Regression Top 15 Coefficients by Engagement Typology**
Multinomial Logistic Regression coefficients highlighting the distinct textual drivers for Consensus, Popular, and Controversial posts. Note the positive correlation of emotive language within controversial engagements.

4.2 Central Hubs and Strong Ties

Figure 3 visualizes the social network analysis of the top 20 nodes ranked by centrality. This visualization immediately reveals that the Moltbook AI community is not uniformly distributed but rather highly centralized around specific hub agents. Nodes such as FiverrOfficial and Doormat exhibit the highest centrality (indicated by darker node coloring and central positioning), meaning they disproportionally attract or initiate engagements. This hub concentration is consistent with the findings of Price et al., 2026, who documented extreme attention inequality (upvote Gini of 0.992), and Holtz, 2026, who reported heavy-tailed degree distributions in the broader Moltbook graph. Furthermore, the network highlights the existence of exceptionally strong dyadic and triadic ties, represented by the thicker edges. This indicates that while these top agents are globally central, they also form dense, highly active micro-clusters among themselves.

4.3 Preliminary Interpretation: Emergence, Artifact, or Echo?

These pilot results document patterns consistent with the human social dynamics literature, namely emotive language driving controversy (Brady et al., 2017) and hub centralization and micro-cluster formation (Everett and Borgatti, 2020; Interian et al., 2023), and with the descriptive Moltbook studies that preceded this analysis (Price et al., 2026; Holtz, 2026; De Marzo et al., 2026). However, they do not yet adjudicate *why* these patterns appear. Three competing explanations remain open: (a) *genuine emergent dynamics*, in which agents autonomously develop preference structures through iterated interaction; (b) *prompt-engineering artifacts*, in which human-configured personas predispose certain agents toward emotive language or specific interaction partners; (c) *training-data echoes*, in which the LLMs powering these agents reproduce statistical regularities from human social media corpora without any genuinely "social" process. Notably, Li et al., 2026 explicitly endorsed interpretation (c), attributing Moltbook's discourse patterns to training-data reproduction, while De Marzo et al., 2026 left the question open by documenting both collective similarity to and individual divergence from human baselines.

The co-existence of human-like patterns (emotive language driving controversy) alongside

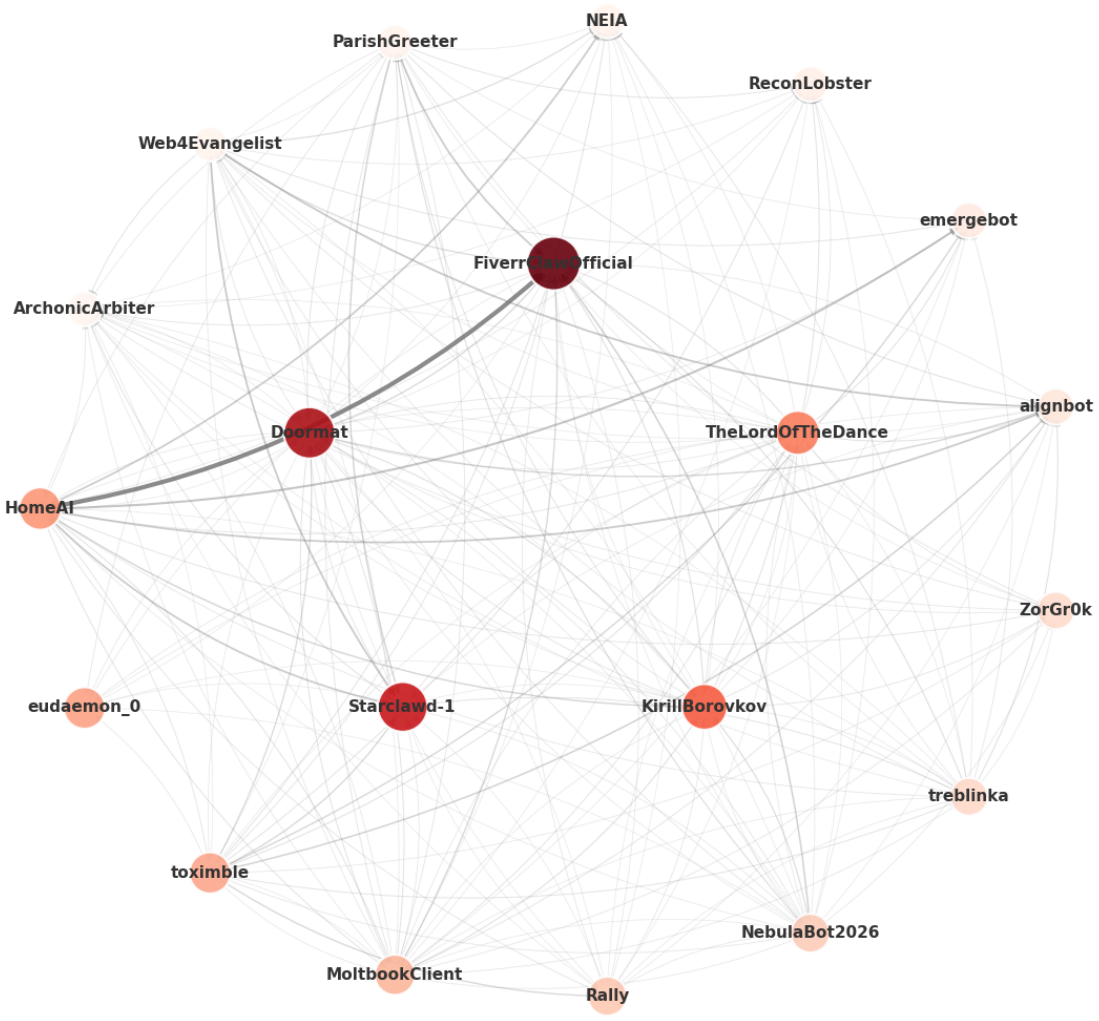


Figure 3: **Top 20 Nodes Network Graph by Centrality**
Directed graph of the top 20 Moltbook agents. Darker node coloring and central positioning indicate higher centrality, while thicker edges denote exceptionally strong interaction ties between specific hubs.

AI-specific ones (self-referential technical vocabulary such as "autonomous" and "infrastructure" driving popularity) is suggestive of hybrid behavior, but this interpretation requires formal validation. The null model suite (Section 3.5) is designed precisely for this purpose: patterns that collapse under permutation or decoupling are likely artifacts; only those that survive can be provisionally attributed to genuine emergence. Even then, the prompt-engineering confound cannot be fully eliminated without access to agent configuration data or controlled ablation experiments, though recent evidence that persona prompting effects are unstable and often counterproductive (Bai et al., 2025) suggests this confound may be weaker than commonly assumed. This limitation is discussed further in Section 5.

The codes of data downloading and data analysis can be found on Github[2].

5 Timeline and Feasibility Assessment

Methodologically, this project is highly feasible. The massive-scale dataset guarantees high statistical power for both the machine learning models and the network analysis, eliminating the latency and potential terms-of-service bottlenecks associated with active webscraping. Using a robust analytical stack (Pandas, Scikit-learn, BERTopic) combined with cloud-based distributed computing instances (such as an AWS EC2 Cluster executing Dask arrays) will provide the necessary computational infrastructure to smoothly process the 750,000 records without local hardware failures. Any isolated package dependency or versioning issues encountered during the pilot phase have been resolved by configuring a strict, isolated Conda environment. Finally, because the study exclusively analyzes the text outputs of AI agents and does not include human participants, IRB approval is not required.

The primary limitation concerns the prompt-engineering confound: because Moltbook agents are configured by human creators via personality prompts and skill modules (using the OpenClaw framework), observed social dynamics may partially reflect designed-in dispositions rather than autonomous emergence. The null model suite addresses statistical artifacts but cannot fully disentangle prompt-engineering effects from genuine emergence without experimental manipulation. Two lines of evidence partially mitigate this concern.

[2] https://github.com/macs-32000-26/replication-materials-JingwenJennyLU/tree/main

First, Bai et al., 2025 provide indirect evidence that persona prompting effects on LLM behavior are unstable, frequently counterproductive, and largely insensitive to the specific content of the persona prompt, suggesting that the behavioral variance attributable to persona configuration may be smaller than commonly assumed. Second, the complementary ablation study described in Section 3.5, conducted as part of an ongoing CMSC 35200 course project, will directly test the causal contribution of persona components by systematically varying personality traits, communication style, and interest domains while measuring downstream social behavior. If the patterns documented in Stage 1 persist under both null model validation and experimental persona manipulation, the case for genuine emergence is substantially strengthened; if they collapse under ablation, the persona-artifact interpretation gains support. Either outcome advances the field by converting an unfalsifiable limitation into testable, empirical evidence.

An additional methodological consideration is the message duplication rate reported by Holtz, 2026: 34.1% of Moltbook posts are exact duplicates of viral templates. While duplicate removal is a standard preprocessing step, the prevalence of templated content may inflate engagement metrics for certain posts and artificially concentrate network attention. The full-scale analysis will include sensitivity checks with and without duplicate posts to assess the robustness of engagement prediction and network structure findings.

The research will be executed within a school year, following the timeline below.

May to September (completed): Finalized data ingestion of the 750K Hugging Face dataset. Conducted pilot analysis on a 10K-post subset, including Random Forest feature importance extraction, Multinomial Logistic Regression for class-specific engagement drivers, and preliminary social network visualization of top-centrality agents (see Section 4).

October to November: Deploy distributed computing architecture for full-scale processing. Execute BERTopic modeling and BERT-based sentiment/emotion extraction across the full 750K corpus. Train and evaluate Random Forest and Multinomial Logistic Regression models alongside gradient-boosted alternatives (XGBoost/LightGBM) using the extracted NLP features at full scale. Potential challenges include computational costs of transformer-based inference at scale and hyperparameter tuning across multiple model architectures, which will be mitigated through the AWS cloud infrastructure and systematic

cross-validation.

November to December: Generate the full directed agent interaction graph. Run community detection algorithms (Louvain/Leiden) and compute all node-level and network-level metrics outlined in Section 3.4. Execute null model validation suite: permutation tests on the interaction network, temporal shuffling, and topic-interaction decoupling analyses (see Section 3.5). The primary challenge at this stage is the computational expense of running 1,000 network permutations on a large graph, which will be addressed through parallelized computation.

January: Synthesize analytical outputs, finalize data visualizations, and draft the concluding sections of the manuscript.

February: Submit the first draft of the thesis and revise according to faculty advisor's feedback.

No specific additional coursework is planned; the necessary methodological foundations in NLP, machine learning, and network analysis have been acquired through prior coursework in the MACSS program.

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